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REPORT



The Dual Role of Motor Abundance in Supporting Upper Limb Performance and Gaze Behavior After Stroke

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ABSTRACT

This study investigated the relationship between motor abundance, gaze behavior, and upper limb performance in individuals post-stroke. Twenty-nine adults with stroke performed an upper limb object transportation task in virtual reality under different visual-motor control conditions. Uncontrolled Manifold analysis quantified two types of variability in upper limb and trunk kinematics: variability that stabilizes a critical performance variable—reflecting use of motor abundance—and variability that disrupts performance (nonfunctional variability). Head orientation relative to the end-effector and task environment was used as an indicator of participants' gaze behavior. Results showed that greater use of motor abundance predicted better task performance with the paretic limb. A similar relationship was observed for the non-paretic limb, but only when nonfunctional variability was low; when it was high, greater motor abundance predicted worse performance. Greater use of motor abundance was also related to gaze patterns that were more variable, more temporally structured, and more oriented to the task environment. Together, these findings point to a dual role of motor abundance in supporting the simultaneous demands of active information sampling and stabilization of task-relevant variables during functional activities.

Introduction

More than 65% of stroke survivors do not regain the ability to effectively move their arms and hands to reach, grasp, manipulate, or transport objects (Levin et al., 2019). These challenges are associated with restricted social participation and a reduced likelihood of leading productive and fulfilling lives (de Graaf et al., 2018; Gadidi et al., 2011). Although a growing number of interventions aim to improve upper limb function post-stroke, many share a common focus on 'normalizing' movement (Aprile et al., 2020; Cavanaugh et al., 2017; Ellis et al., 2018; Franceschini et al., 2012; Turolla

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et al., 2013; van Dijk et al., 2017). The underlying—though often unarticulated assumption—is that the movements of non-disabled individuals represent the ideal reference, and that any deviations from this norm are errors requiring correction (Gibson, 2016; Roush & Sharby, 2011).

This assumption conflicts with the lived experiences of individuals with disabilities, who report that everyday environments often require them to move their bodies in ways that depart from the norm (Dokumaci, 2013, 2017, 2023; Rahlin et al., 2019; Sobchack, 2017). Experimental findings echo these experiences: individuals post-stroke frequently increase involvement of the trunk or shoulder and rely less on fine distal control compared to nondisabled controls (Cirstea & Levin, 2000; Levin et al., 2002, 2009; Reisman & Scholz, 2006; Roby-Brami et al., 2003). Grounded in basic findings from motor control research, reviewed below, we hypothesized that deviations from normative movement after stroke may reflect not only impairment, but also functional reorganization that enables task success within the constraints and opportunities defined by the individual's available capabilities and environmental supports (see Silva & Schwab, 2024 for more details).

Findings from basic research in motor control

Converging evidence shows that successful motor performance does not require individuals to consistently produce an idealized 'normal' or 'correct' movement pattern. Instead, individuals succeed by organizing subsets of motor elements (i.e. joints and muscles actuating them) in ways that respond to contextual and task-specific demands (Steenbergen et al., 1995; van der Kamp & Steenbergen, 1999; van der Steen & Bongers, 2011). This is possible because the motor system provides more degrees of freedom than a task strictly requires—a principle known as motor abundance (Latash, 2000; 2012). Studies show that non-disabled individuals exploit this abundance by varying joint configurations across repeated successful attempts of everyday tasks, such as standing up from a chair (Andrade et al., 2023; Scholz & Schöner, 1999) and reaching toward targets (Mattos et al., 2011). In other words, skilled individuals exhibit structured variation in movement patterns that reflects motor flexibility: they use of a range of movement strategies that preserve task-relevant outcomes, such as maintaining a consistent trajectory of the center of mass during sit-to-stand transitions or of the hand during reaching.

This flexibility in motor coordination is not unique to non-disabled individuals; rather, it reflects a broader principle of how movement can be adapted to task demands. For example, when children with cerebral palsy perform reciprocal aiming tasks with a pointer, they rely primarily on distal joints (wrist and elbow), similar to age-matched non-disabled peers—but only when the targets are relatively large (Figueiredo et al., 2015). As accuracy demands increase, they reduce movement at the wrist and elbow and increase motion at the shoulder and trunk. Rather than indicating dysfunction, this shift can be interpreted as a functionally appropriate adaptation. Given their reduced ability to control fine movements at distal joints—especially under high-precision demands—freezing distal joints and compensating with proximal segments serves as an effective strategy. Notably, non-disabled individuals exhibit similar adjustments under conditions requiring increased accuracy (Steenbergen et al., 1995; van der Kamp

& Steenbergen, 1999), suggesting that such changes are not impairment-specific, but are instead a marker of flexibility and adaptability (Figueiredo et al., 2015; Fonseca et al., 2004).

Together, these findings support a growing consensus in motor control research: variability in motor performance is not (always) an error requiring correction, but may be an expression of the use of motor abundance—a prerequisite for successful task performance across variable circumstances (Cavanaugh et al., 2017; Hadders-Algra, 2010; Latash et al., 2002; Todorov & Jordan, 2002). From this perspective, persistent upper limb difficulties following stroke may not stem from an inability to move in normative ways, but rather a diminished capacity to explore available motor abundance during everyday activities. The present study offers an initial empirical evaluation of this possibility by investigating whether post-stroke functional difficulties are associated with reduced use of motor abundance.

The uncontrolled manifold: a tool to quantify the use of motor abundance

The Uncontrolled Manifold (UCM) is a framework and analytical tool that can be used to examine the role of motor abundance in upper limb function post-stroke (Scholz & Schöner, 1999; reviewed in Latash et al., 2007). The UCM framework is based on the assumption that motor abundance allows for the stabilization of important performance variables while enabling the flexibility to accommodate task and contextual demands (Latash, 2012; Latash et al., 2007). Importantly, the UCM framework recognizes that the use of motor abundance is not simply equivalent to the total amount of variability in the state of motor elements supporting task performance; after all, not all movement variations lead to task success. For instance, in a reaching task, any combination of joint motions that position the hand on the target supports successful performance. However, if variations in the position of one joint are not appropriately met with reciprocal variations in another joint (e.g. when a reduction in elbow extension is not compensated by increased trunk flexion), the hand may fail to reach the target.

Consistent with this understanding, UCM analysis partitions variability in the states of motor elements (e.g. movements of body segments) across multiple performance attempts into functional and nonfunctional components, referred to as V_{UCM} and V_{ORT} respectively (see Latash, 2012; Latash et al., 2007). V_{UCM} is a measure of co-variation among candidate motor elements that stabilizes a critical performance variable (e.g. terminal hand position required for successful performance of a reaching task). V_{UCM} can, thus, be used to quantify the use of motor abundance and evaluate its relationship to upper limb performance in individuals post-stroke. In contrast, V_{ORT} represents variations that do not stabilize the performance variable and would disrupt performance (thus, V_{ORT} is considered nonfunctional). UCM analysis also provides a synergy index (ΔV), which reflects the normalized difference between V_{UCM} and V_{ORT} . When V_{UCM} is greater than V_{ORT} , one can conclude that the motor elements are organized into a synergy that stabilizes the performance variable under analysis.

Using the UCM framework, prior work has demonstrated that individuals post-stroke can stabilize hand position during reaching tasks by flexibly coordinating multiple joints—including the trunk—particularly when the target remains within the arm's

functional range (Reisman & Scholz, 2003, 2006; Tomita et al., 2020). However, when reaching beyond arm's length—especially toward the hemiparetic side—this flexibility was reduced, likely reflecting limitations in joint motion and altered shoulder–elbow coordination (Reisman & Scholz, 2003, 2006). Taken together, these findings suggest that when impairments do not preclude *effective* compensatory strategies, individuals post-stroke retain the ability to organize and exploit motor abundance to support task performance. The present study asks whether the extent to which individuals spontaneously use available degrees of freedom relates to their success in performing a functional upper limb task.

On the relation between the use of motor abundance & visual activity

The ecological approach defines visual activity as the process of sampling the optic array for information relevant to the regulation of action (Gibson, 1979). When reaching toward an object, sighted individuals engage in this process by actively orienting their gaze—that is, the eyes, typically in coordination with the head—toward relevant aspects of the environment (Ma-Wyatt & McKee, 2006; Schlicht & Schrater, 2007; Sivak & MacKenzie, 1990). Yet visual activity cannot be reduced to movement of the eyes and head alone, because effective orientation of gaze may also depend on broader bodily adjustments, such as leaning forward or walking to reposition the body relative to the object. Crucially for present purposes, some of the same body segments that support visual activity also play a direct role in postural regulation and hand transport.

The trunk is a case in point. Trunk movements may amplify changes in head and, consequently, eye position, compensate for the postural consequences of head motion, or help stabilize the head as a frame of reference for reaching (Bertenthal & Von Hoften, 1998). At the same time, trunk motion directly affects the trajectory of the hand (Reisman & Scholz, 2003, 2006). Thus, coordination among the head, trunk, and upper limb simultaneously supports visual activity, postural control, and hand transport. As this coordination unfolds, changes in posture and limb configuration continuously reshape the visual information available, which in turn further constrains ongoing movement.

In this ongoing perception–action cycle, motor abundance may play a critical role in supporting the simultaneous demands of stabilizing task-relevant variables and the orientation of gaze under evolving constraints. While the contribution of motor abundance to performance stabilization has been widely discussed, its integral role in supporting visual guidance—critical to task execution—remains poorly understood. This gap may be particularly important in the context of stroke, where several common consequences may increase the demands placed on active visual exploration while simultaneously constraining the coordination required to support it.

Three factors, in particular, may shape how individuals post-stroke coordinate visual activity and movement during task performance. First, approximately 60% of stroke survivors experience visual impairments, including visual field loss, eye movement disorders, loss of central vision, and visual inattention/neglect (Falkenberg et al., 2020; Hepworth et al., 2016). Second, stroke survivors often present reduced mobility and impaired postural control (Levin et al., 2009; Schwab et al., 2023), which, as discussed above, contribute to both gaze orientation and the stabilization of task-relevant

variables (Bertenthal & Von Hofsten, 1998). Third, individuals post-stroke tend to shift their gaze toward their own bodies and movements (Bonan et al., 2004; Meadmore et al., 2018)—a pattern that may alter how perceptual-motor exploration unfolds during task performance. In light of these constraints on visual and motor resources, motor abundance may enable individuals post-stroke to meet the concurrent demands of active information sampling and stabilization of task-relevant variables critical for successful upper limb function.

Study objective, specific Aims, and hypotheses

Stroke-related disability is often characterized by marked deviations of movement patterns from the norm. These deviations are typically interpreted as a sign of motor impairment underlying persistent upper limb difficulties post-stroke. In the present study, we examine an alternative hypothesis: that persistent functional difficulties arise from reduced motor flexibility to concurrently stabilize task-relevant variables while supporting visual exploration of the environment. We evaluated this general hypothesis through two specific aims.

The first aim was to determine whether use of motor abundance relates to task performance following stroke. We hypothesized that individuals who exhibit greater use of motor abundance—quantified by higher values of V_{UCM} —would perform better in a goal-directed, object transportation task. We also explored whether this relationship was shaped by task-related factors, such as the specific strategy employed to transport the object to the prescribed location and the phase of the transportation movement. This was based on the idea that different execution demands may influence how the use of motor abundance contributes to successful performance.

The second aim was to examine whether the use of motor abundance is related to gaze behavior. To characterize gaze behavior, we computed a continuous measure of head-gaze angle (describe in detail below), operationalized as head orientation relative to the task environment (McGuckian et al., 2019; Turton et al., 2018; van Andel et al., 2019). Larger head-gaze angles indicate that gaze is oriented farther from the moving effector, toward more distal, task-relevant regions of the environment, whereas smaller angles indicate gaze that remains more closely anchored to the effector. We hypothesized that greater magnitude and variability in head-gaze angle, reflecting greater environmental exploration, would covary with greater use of motor abundance. If the hypotheses associated with Aims 1 and 2 are supported, the study would provide evidence for the proposed dual role of motor abundance in supporting both stabilization of task-relevant variables and gaze behavior during ongoing task performance after stroke.

Methods

Participants

Thirty-seven adults aged 37–82 years previously diagnosed with any type of stroke (>6 months prior to enrollment) volunteered to participate in the study. All gender identities, races, and ethnicities were eligible for participation.

To be included, participants had to demonstrate cognitive capacity to understand and follow task instructions, self-report normal or corrected-to-normal visual acuity, and have no other medical condition affecting upper limb function. Participation was not restricted based on current involvement in rehabilitation programs. Exclusion criteria were severe motor impairment precluding completion of the experimental task (e.g. significant weakness, severe arm spasticity, poor trunk control), inability to understand the task even after practice, limited communication abilities, pain with weight-bearing rated $>4/10$, dizziness or nausea, or additional medical condition that would preclude the valid administration of study measures (e.g. seizure disorders, neurological conditions beyond stroke, spatial neglect).

Ethical approval was obtained from the University of Cincinnati Institutional Review Board, and all participants provided informed consent to participation. Research was conducted in accordance with the principles stated in the Declaration of Helsinki.

Material and apparatus

Participants wore a VIVE virtual reality (VR) headset (HTC Corporation, Bellevue, WA) that displayed a virtual environment (described below) created in Unity (v. 2019; Unity Technologies, San Francisco, CA). The VR scene was rendered at a refresh rate of 90 Hz. Participants interacted with objects in the virtual environment through integration of the Unity platform with a motion analysis system (Motion Analysis Corp. Santa Rosa, CA, USA).

Nineteen Kestrel motion-capture cameras (Motion Analysis Corp. Santa Rosa, CA, USA) recorded 3D kinematics of participants' upper limbs and trunk at a sampling rate of 120 Hz. Twenty retroreflective markers were placed on each participant, as illustrated in [Figure 1](#). Two markers were placed on each hand (on the heads of the second and fifth metacarpals), two on each wrist (on the styloid process of the radius and ulna), two on each elbow (on the medial and lateral humeral epicondyles), one on each shoulder (on the scapular acromion), two on the trunk (on the spinous processes of the seventh cervical and twelfth thoracic vertebrae), and four on the pelvis (two on the highest part of the iliac crest and two on the posterior iliac spine). Eight virtual markers, computed as the midpoint between two retroreflective markers, were generated for the hands (between the markers on the heads of the second and third metacarpals), wrists, elbows, and pelvis (between the markers on the iliac crest and between the markers on the posterior iliac spine).

Virtual environment and experimental task

Participants experienced the virtual environment from a first-person perspective while seated on a chair without back support ([Figure 2a](#)). In the VR environment, they viewed a table in front of them, a goal container (20.5 cm in length, 7 cm in width, and 7 cm in height) located farther ahead, and a narrow bridge (10 cm wide) connecting the two ([Figure 2b](#)). The height of the virtual table was adjusted to each participant's most comfortable height, and the length of the bridge was set to 70% of their functional arm's length.



Figure 1. Front and back views of retroreflective markers placed on participant's upper limbs and trunk.

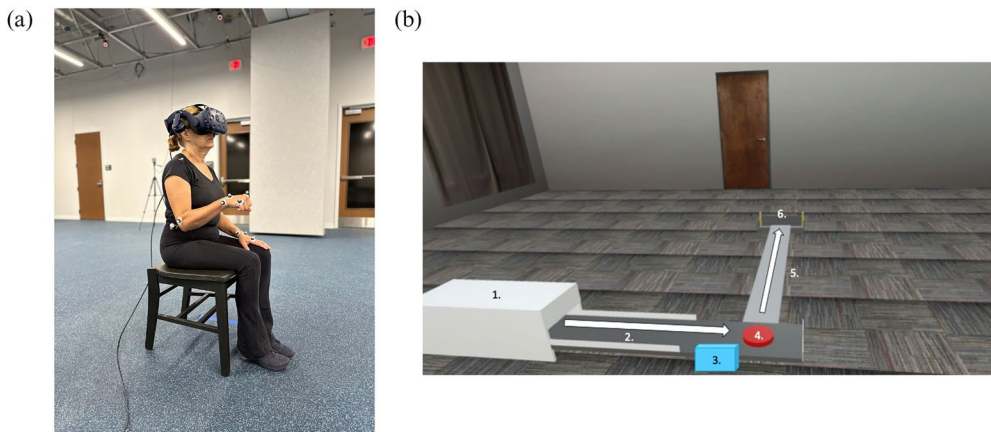


Figure 2. (a) Experimental setup. (b) Transportation task in virtual reality. Arrows indicate the trajectory of the puck from the moment of release until it is directed to the goal container. 1. Puck dispenser, 2. Table, 3. Virtual pad, 4. Puck, 5. Bridge, 6. Goal container.

The objective of the task was to transport pucks (8 cm in diameter) from the table, across the narrow bridge, and into the goal container using one upper limb at a time. A virtual puck dispenser positioned at the edge of the table presented pucks to participants one at a time. Participants were instructed to use a virtual pad (15 cm in length, 7 cm in height, and 2 cm in thickness)—controlled by their hand

movements—to transport the pucks across the bridge into the goal container. Friction between pucks and the surfaces was eliminated, allowing the pucks to slide freely across the bridge (similar to air-hockey). Pucks that fell off the bridge became unusable.

Participants were required to make as many attempts as required to successfully transport 20 pucks into the goal container using two pre-determined task strategies: pushing or hitting. In the *pushing* condition, participants were instructed to use the virtual pad to continuously push the puck across the entire length of the bridge, releasing it only when it was as close as possible to the goal container. In the *hitting* condition, participants were instructed to use the virtual pad to strike the puck from its initial position on the table and let it slide across the bridge. A video demonstrating a nondisabled individual performing the pushing condition is available at <https://osf.io/8kewm/overview>.

This experimental task was designed to simulate a common challenge experienced by individuals with stroke-related disability: the need to make multiple attempts before achieving a successful outcome. Rather than using a simplified reaching task to mitigate task failures, as in previous studies involving participants with motor disabilities (Gera et al., 2016a, 2016b), we selected a task in which performance could be evaluated based on the number of attempts required to achieve a set number of successful outcomes (see Data Analysis & Measures).

Procedure

First, participants completed the Patient-Reported Outcomes Measurement Information System (PROMIS) Physical Function with Mobility Aid Short Form (v1.0) and Upper Extremity Short Form (v2.1—7a), which assess perceived functional status (Hung et al., 2017; Rose et al., 2008, 2014). Higher scores on both forms indicate better physical functioning.

Participants then completed practice trials to broadly familiarize themselves with the experimental setup and task. Once familiarization was complete, participants received explicit instruction and brief practice on the two pre-determined transportation strategies that defined the experimental conditions of the task: *pushing* and *hitting*. The order of these experimental conditions was counterbalanced across participants. For each condition, participants always used their non-paretic limb first, followed by their paretic limb.

Pucks were delivered one at a time at the participant's command (that is, the task was self-paced and no time constraints were imposed) until 20 successful trials were completed for each condition with each limb—a number chosen to meet the requirement of UCM analysis (Freitas et al., 2019). A trial was considered successful if: (a) the participant used the instructed transportation strategy—pushing or hitting—confirmed by direct observation; and (b) the puck landed in the goal container. Between attempts, participants were instructed to return their upper limb to the initial position before the next puck was presented. A mandatory one-minute rest period was provided between condition blocks to prevent fatigue. Participants could request additional breaks as needed. Sessions lasted between 1 and 1.5 h.

Data analysis & measures

Upper limb task performance

The primary performance metric was the number of attempts required to complete 20 successful trials for each condition (pushing and hitting) and each limb (paretic and non-paretic). Fewer attempts indicate better (more consistent) performance. While coarse, this measure captures a functionally meaningful aspect of performance in the present task, namely, the need to make repeated attempts to achieve a successful outcome, which is a common challenge in everyday action after stroke.

UCM analysis

UCM analysis was conducted separately for the pushing and hitting conditions. This allowed trials to be treated as instances of the same underlying movement pattern, enabling the required time normalization of the movement trajectories. The analysis proceeded in three main steps, briefly described below.

Step 1: defining performance and elemental variables. Performance and elemental variables were computed from motion capture data after all marker position time-series were low-pass filtered using a fourth-order Butterworth filter with a 6 Hz cutoff. The performance variable was the 3D hand position, expressed in Cartesian coordinates, representing the task-relevant outcome to be stabilized during puck transport.

Elemental variables were defined as joint angles describing the 3D configuration of five body segments: lower trunk, upper trunk, shoulder, elbow, and wrist. Each joint was represented by a single scalar angle, computed within a local plane defined by the instantaneous three-dimensional configuration of adjacent segment markers. Specifically, for each joint, a plane was defined using three anatomically relevant markers (e.g. for the elbow: upper arm, elbow, and forearm markers), and the joint angle was calculated as the angle between the two segments projected onto this plane. Because the orientation of this plane varies continuously with the configuration of the limb, rotations outside the anatomical sagittal plane influenced the computed joint angles and were incorporated into the elemental variables. This approach preserves a single-angle representation per joint while allowing three-dimensional deviations in movement to be reflected in the analysis.

Compared to multi-degree-of-freedom representation of joint motion, this formulation captures out-of-plane variability without increasing the dimensionality of the elemental-variable space. In addition, because joint motion is not decomposed into multiple rotational components, it avoids dependence on rotation-order conventions and yields a more stable estimation of the Jacobian, reducing susceptibility to noise amplification. The tradeoff of this approach is that the absolute values of the angles do not map directly onto standard anatomical components of joint motion (e.g. flexion-extension or internal-external rotation), which are defined with respect to fixed coordinate systems.

To provide a visual impression of the kinematic organization captured by this formulation, we included joint-angle trajectories of three participants performing each task condition with their paretic (right) limb (see [Figure 3](#)). The figure displays time-normalized joint angles for the upper limb and trunk segments across the course

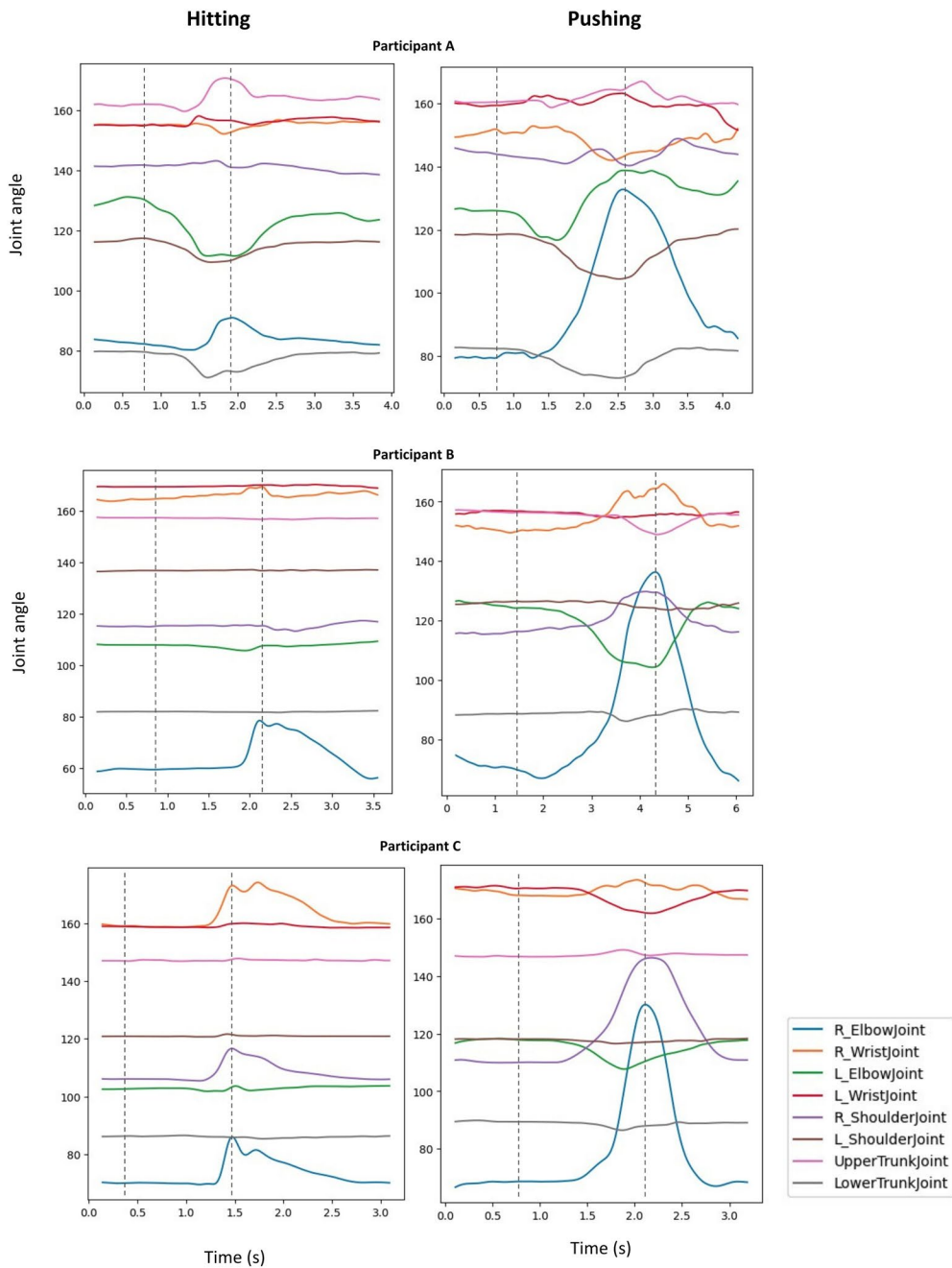


Figure 3. Trial-level joint-angle trajectories illustrating the magnitude and pattern of change over time for the upper limb and trunk segments from three different participants performing the task with their paretic (right) limb. Vertical dashed lines indicate movement onset and termination. Joint angles are computed within local, configuration-dependent planes and therefore reflect the relative orientation between adjacent segments in three-dimensional space rather than conventional anatomical rotations. Differences across conditions (hitting and pushing) and participants highlight how the relative contributions of segments are shaped by task demands.

of a trial. Because each angle reflects the relative orientation between adjacent segments within a configuration-dependent plane, these trajectories illustrate the magnitude and pattern of change in segment orientations over time, rather than conventional anatomical rotations. These trajectories provide an intuitive depiction of how segment orientations evolve over time and how the relative contributions of segments are shaped by task conditions. For example, in the pushing condition, participants tended to use greater elbow and shoulder movements (blue and purple curves, respectively), whereas these joints showed relatively lower contribution to the hitting condition.

Step 2: identifying movement onset and termination. An automated algorithm was used to identify the onset and termination of each transportation movement. Movement onset was defined as the first velocity peak in any of the three coordinates of the virtual marker on the hand, corresponding to the initiation of the participant's forward movement to transport the puck. Movement termination was defined as the point of maximum forward displacement of the hand marker, typically corresponding to peak elbow extension and puck release. In approximately 5% of trials, the automated method failed to identify onset and/or termination; in these cases, movement boundaries were determined through visual inspection. For each included trial, the portion from movement onset to termination was normalized to 100 samples in evenly spaced steps with cubic polynomial interpolation. Movement times (MTs) exceeded 1 s across all conditions (see Results). At the sampling resolution used (120 Hz), this ensured that each trial contained at least 100 observed data points, such that normalization to 100 time points did not require extrapolation beyond the recorded movement interval.

Step 3: projecting deviations and computing variability. A geometric model was used to describe the relationship between the elemental variables and the performance variable (see Figure 4), and the Jacobian matrix of this model was computed. This matrix was used to define two orthogonal subspaces. The UCM is the null space of the Jacobian and contains variations in joint configurations that preserve hand position. The orthogonal complement of the null space (ORT) contains variations that affect hand position. For each time frame of the movement, deviations of elemental variables from their trial-averaged trajectories were computed and projected onto the UCM and ORT subspaces.

Two measures of variability were derived from these projections (see Solnik et al., 2013, for further detail). V_{UCM} was defined as the average of the squared projections onto the UCM subspace and reflects functional variability—co-variation among elements that stabilize the performance variable and indicates use of motor abundance. V_{ORT} was defined as the average of the squared projections onto the ORT subspace and represents nonfunctional variability—variability that disrupts the stability of the performance variable (here, the hand position). Because the dimensionality of the UCM and ORT subspaces differs, both V_{UCM} and V_{ORT} were normalized by the number of dimensions in their respective subspaces, allowing for meaningful comparison.

V_{UCM} , V_{ORT} , and ΔV were computed at each time step throughout the transportation movement, from onset to termination, yielding a frame-by-frame account of motor coordination for each trial. The resulting time-normalized series were divided into two phases (early, first half; and late, second half). For each phase, average V_{UCM} ,

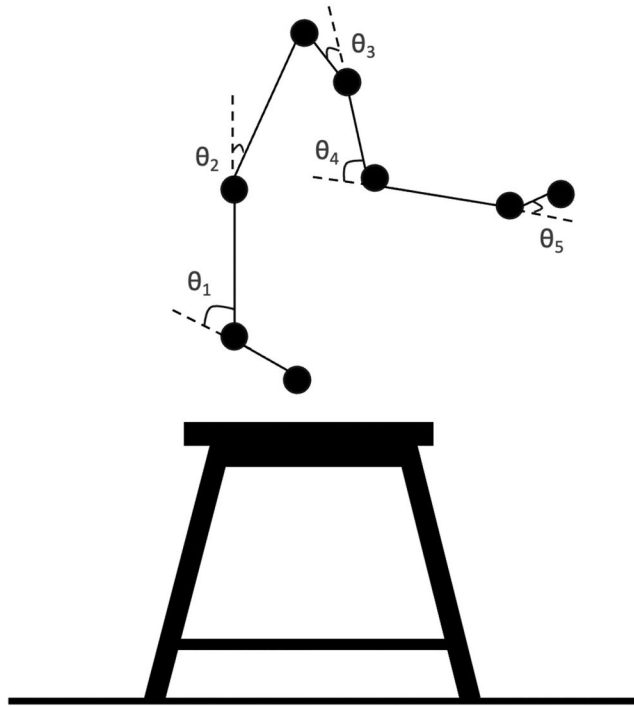


Figure 4. Schematic representation of geometric model linking joint angles to changes in the performance variable. The five angles used in the Jacobian matrix (θ_1 – θ_5) correspond to the lower trunk, upper trunk, shoulder, elbow, and wrist. For visualization purposes, the model is depicted in a planar configuration; however, in the present analysis, each angle is computed within a local, configuration-dependent plane defined by the three-dimensional arrangement of adjacent segment markers. Thus, the angles represent the relative orientation between segments in three-dimensional space rather than fixed anatomical rotations.

V_{ORT} and ΔV (normalized difference between V_{UCM} and V_{ORT}) values were computed and used in subsequent statistical analyses.

Measures of gaze behavior: head-gaze angle

Head movements relative to the environmental scene were used as an indicator of gaze behavior (McGuckian et al., 2019; Turton et al., 2018; van Andel et al., 2019). We computed a metric—head-gaze angle—to quantify the extent to which participants changed head orientation relative to the task environment, using the virtual pad (proximal to and controlled by the hand) as a reference point.

The head-gaze angle was defined as the smallest angle between (a) a vector originating at the center of the head and pointing toward the center of the virtual pad, and (b) a vector also originating at the center of the head but extending outward in the direction of head orientation, as determined by ray-casting—a method that projects an invisible line forward from the headset to estimate where the participant is looking within the virtual environment (Figure 5). Head position and orientation were continuously tracked *via* the VR headset's built in motion sensors. Head-gaze angle data were recorded from the beginning of each trial (when pucks first appeared in the

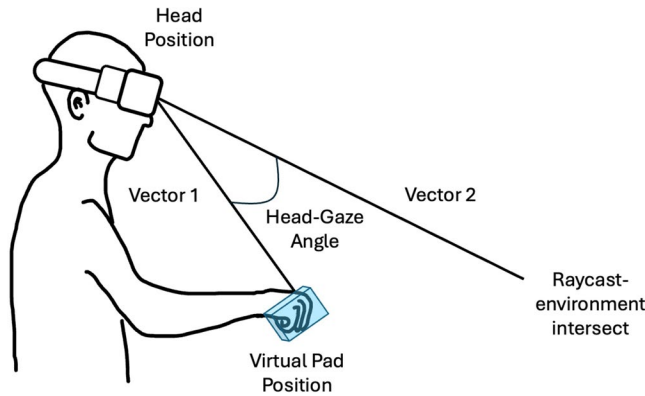


Figure 5. Head-gaze angle is the smallest angle between a vector that originates in the head and ends in the center of the virtual pad (Vector 1) and a vector that originates in the head and extends toward the environment (Vector 2).

scene) until the moment of transportation movement termination, as defined in Step 2 in the UCM analysis sub-section.

From the head-gaze angle time-series, we computed three summary metrics: *mean*, *range*, and *fluctuation index*. The magnitude of the head-gaze angle reflects the degree to which participants oriented their head—and given expectation of head-eye coupling their gaze—away from the virtual pad and toward more distal features of the task environment. A larger mean angle indicates that gaze was directed further into the environment while a smaller mean angle reflects a tendency to look more closely to the end-effector (i.e. the virtual pad). The range of head-gaze angle was calculated as the difference between maximal and minimal values observed during the trial, reflecting the extent to which participants varied head orientation relative to the end-effector while performing the task.

The fluctuation index captured the temporal dynamics of head-gaze behavior and was computed using custom MATLAB scripts (R2022a, MathWorks, Natick, MA) based on procedures described by Schiepek and Strunk (2010), validated for short time-series. The analysis involved three steps. First, the head-gaze angle time series was segmented at *k-boundaries*, which mark transitions between behavioral trends classified as increasing, decreasing, or stable (see Figure 6 for an illustration). An increasing trend indicates that the head and virtual pad are moving away from each other; a decreasing trend indicates they are moving closer; and no change indicates a stable relationship. Second, for each segment, the rate of change was calculated as the absolute difference in head-gaze angle between consecutive *k-boundaries* divided by the number of data points in the segment. Third, the fluctuation index was computed by summing these rate-of-change values across all segments. Higher values indicate more rapid and frequent shifts in head orientation behavior, indicating a more richly varied pattern of visual exploration.

Statistical analysis

Distributions of study variables, stratified by experimental condition and limb, were examined to assess normality. V_{UCM} , V_{ORT} , and fluctuation index of head-gaze angle

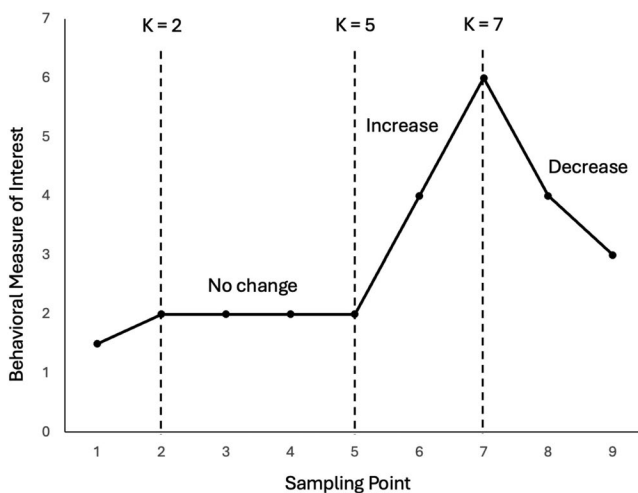


Figure 6. Illustration of the behavioral trends that can be found in head-gaze angle time series.

violated the normality assumption and were log-transformed. These transformations substantially reduced observed skewness, bringing distributions closer to normal for inferential statistics. ΔV was normalized using modified Fisher's z-transformation (Solnik et al., 2013). The performance variable (number of attempts) was positively skewed, reflecting its bounded nature and was therefore not transformed. Instead, model assumptions were evaluated at the level of residuals. Residual diagnostics indicated no meaningful departures from normality or homoscedasticity for models including performance as the dependent variable; provided analyses were conducted separately for non-paretic and paretic limbs.

A series of preliminary analyses was conducted to characterize participants' overall task performance and UCM-derived measures. Subsequent analyses were organized around the study's specific hypotheses. For clarity and interpretability, each analysis will be introduced in the corresponding section of the Results.

Most of these analyses were conducted using linear mixed-effects models, implemented in R via the *lme4* package, to accommodate the hierarchical structure of the data. A backward stepwise approach was used to build these models, as outlined by West et al. (2015). Starting with a full model, nonsignificant effects were removed sequentially, beginning with the highest-order interactions. At each step, model fit was evaluated using the deviance statistic ($-2 \log$ -likelihood, or $-2LL$), comparing nested models with and without the effect of interest. Changes in $-2LL$ follow a chi-square distribution, with degrees of freedom equal to the difference in the number of estimated parameters, enabling formal significance testing.

Final models retained all statistically significant higher-order interactions, along with their associated lower-order terms and main effects. For models with log-transformed outcome variables, coefficients were exponentiated ($\exp(\beta)$) and interpreted as multiplicative effects on the original outcome scale. For models with log-transformed predictor variables, we reported both the raw regression coefficients and the estimated change in the outcome associated with a 10% increase in the predictor. This was calculated by multiplying the regression coefficient by $\log(1.10)$, which corresponds to

a 10% increase on the log scale. These transformations allow effect sizes to be interpreted directly in terms of proportional changes in the outcome variable, providing a meaningful index of effect magnitude on the original measurement scale. These values are reported in the Results and reflected in model summary tables to support interpretation.

Results

Of the 37 individuals initially enrolled in the study, four were unable to complete the experimental task due to significant motor impairment, two were unable to understand the task instructions, and technical issues precluded task completion for an additional two participants. Thus, the final sample included 29 participants who completed at least one experimental condition and were included in subsequent analyses—see Table 1 for the demographics of the final sample.

All included participants successfully completed the *hitting* condition using both their paretic and non-paretic upper limbs. However, seven participants were unable to complete the *pushing* condition with their non-paretic limb. This finding suggests that the push strategy may require physical resources—such as sufficient muscular strength to sustain longer arm movements or greater range of motion—that are not necessary in the hitting condition.

A subset of trials was excluded during preprocessing. Out of 2320 trials, 168 were removed due to poor data quality identified during visual inspection (e.g. marker drop-out or signal noise) and three were excluded because movement onset could not be reliably identified. Following these exclusions, all remaining participants had a sufficient number of valid trials to conduct UCM analysis in each condition and limb. The UCM analysis was performed using an average of 18 trials (range: 16 to 20) per participant per condition and upper limb.

Preliminary analyses

As noted previously, to provide context for interpreting the results of the primary hypothesis tests, we first conducted a series of analyses to characterize overall task performance and UCM metrics, as well as their relationships to limb impairment and task condition.

Upper limb task performance

A linear mixed-effects model with participant as a random effect was used to examine task performance—defined as the number of attempts required to complete 20 successful trials—varied by limb (paretic vs. non-paretic), task condition (hitting vs. pushing), and their interaction. The model revealed a significant main effect of limb, $t(75.29) = 7.72$, $p < .001$. On average, participants required more attempts when using their paretic limb ($M = 27.14$, $SD = 11.07$) compared to their non-paretic limb ($M = 22.12$, $SD = 2.90$), indicating reduced efficiency of movement execution.

This effect was qualified by a significant limb $\times$ condition interaction, $t(77.35) = 5.66$, $p = .03$. Post-hoc comparisons with Tukey adjustment revealed a significant

Table 1. Final sample demographics.

Characteristic	<i>n</i> = 29
Age, y (Mean ± SD)	60.80 ± 10.80
Gender, n (%)	
Female	13 (44.83)
Male	16 (55.17)
Race, n (%)	
Black	5 (17.24)
Caucasian	24 (82.76)
Height, m (Mean ± SD)	1.73 ± 0.10
Weight, Kg (Mean ± SD)	86.05 ± 20.53
Stroke Age, mo. (Mean ± SD)	
Stroke Type, n (%)	
Hemorrhagic	9 (31.03)
Ischemic	19 (65.52)
Hemorrhagic + Ischemic	1 (3.45)
Paretic Side, n (%)	
Left	17 (58.62)
Right	12 (41.38)
Mini-Mental Status Exam (Mean ± SD)	29.07 ± 1.67
PROMIS Physical Function with Mobility Aid Short Form ((Mean ± SD)	47.97 ± 5.94
PROMIS Upper Extremity Short Form ((Mean ± SD)	21.57 ± 8.83
Hours/Week in Physical or Occupational Therapy, n (%)	
No therapy	23 (79.31)
<1 h	1 (3.45)
1-2 h	3 (10.34)
2-4 h	2 (6.90)
Highest Level of Education, n (%)	
High school diploma	2 (6.90)
Some college	3 (10.34)
Bachelor's degree	20 (68.97)
Graduate degree	4 (13.79)
Median Annual Household Income, n (%)	
<\$20,000	3 (10.34)
\$20,000-\$40,000	2 (6.90)
\$40,000-\$60,000	8 (27.59)
\$60,000-\$80,000	3 (10.34)
\$80,000-\$100,000	3 (10.34)
>\$100,000	9 (30.03)
Prefer not to say	1 (3.45)
Other Medical Conditions (n)	Anxiety (1); Arteriovenous malformation (2); Arthritis (2); Crohn's disease (1); Depression (3); Diabetes (5); Grave's disease (1); Heart attack (2); Hypertension (12); Low back pain (2); Mixed connective tissue disease (1)

Abbreviations: PROMIS – Patient-Reported Outcomes Measures Information System (Physical Function: 55 maximum score; Upper Extremity: 35 maximum score).

performance difference between limbs in the hitting condition ($M_{\text{diff}} = 7.72$, 95% CI [3.06, 12.389], $p < .0001$), but not in the pushing condition ($M_{\text{diff}} = 2.06$, 95% CI [3.01, 7.14], $p = .71$; see [Figure 7](#)). No significant main effect of condition was observed, $t(75.29) = -0.17$, $p = .92$. In sum, the limb-related performance deficit was primarily evident in the hitting condition, which appears to have placed greater demands on motor execution among those able to perform both tasks than the pushing condition.

Movement times (MTs) were examined to characterize the temporal structure of task performance. A linear mixed-effects model revealed that MTs were significantly longer in the pushing than in the hitting conditions, $t(80.16) = 9.37$, $p < .001$, and significantly longer for the paretic than the non-paretic limb, $t(80.16) = 2.06$, $p = .043$.

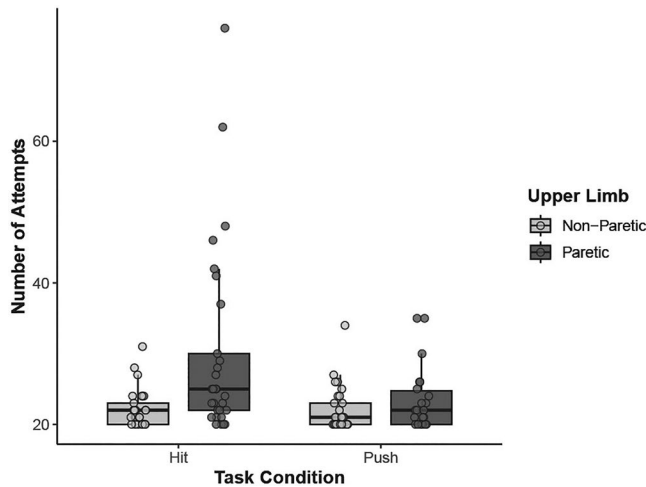


Figure 7. Task performance by experimental condition and upper limb. Number of task attempts (i.e. worse performance) was higher when using the paretic limb than the non-paretic limb. Differences in performance between upper limbs were attenuated for the pushing condition.

The interaction between condition and limb was not significant, $t(80.890) = -0.23$, $p = .82$. Mean MTs were 1.80 s, 95% CI [1.66, 1.94], for hitting with the non-paretic limb; 2.13 s, 95% CI [1.79, 2.46], for hitting with the paretic limb; 3.42 s, 95% CI [2.84, 4.00], for pushing with the non-paretic limb; and 4.04 s, 95% CI [3.16, 4.91], for pushing with the paretic limb.

Relationship between task performance and self-reported functional status

Simple linear regressions were conducted to determine whether task performance with the paretic limb predicted self-reported functional status. No significant relationships were found between task performance and scores on the PROMIS Physical Function short form in either the pushing condition, $F(1, 20) = 2.50$, $p = .13$, $R^2 = .11$, or the hitting condition, $F(1, 27) = 0.47$, $p = .50$, $R^2 = .02$. These findings suggest that performance in the experimental task may not align closely with measures of *general* physical functioning.

In contrast, performance was significantly associated with self-reported upper limb function as measured by the *PROMIS Upper Extremity* short form. In the pushing and hitting conditions, poorer performance (i.e. more attempts) was significantly related to lower PROMIS Upper Extremity scores, $F(1, 20) = 10.39$, $p = .004$, $R^2 = .34$ and $F(1, 27) = 12.44$, $p = .002$, $R^2 = .32$, respectively (see Figure 8). To account for the skewed distribution of the performance variable, we conducted complementary non-parametric analyses. Spearman rank correlations revealed consistent associations in both conditions (Hitting: $\rho = -0.54$, $p = .002$; Pushing: $\rho = -0.57$, $p = .005$), indicating that the observed relationships were robust and not driven by distributional assumptions or a small number of influential observations. In all cases, higher self-reported functional ability was associated with better task performance.

We also evaluated MT as an alternative index of task performance. Although MT differed across task conditions and limbs, neither the mean nor variability (SD) of

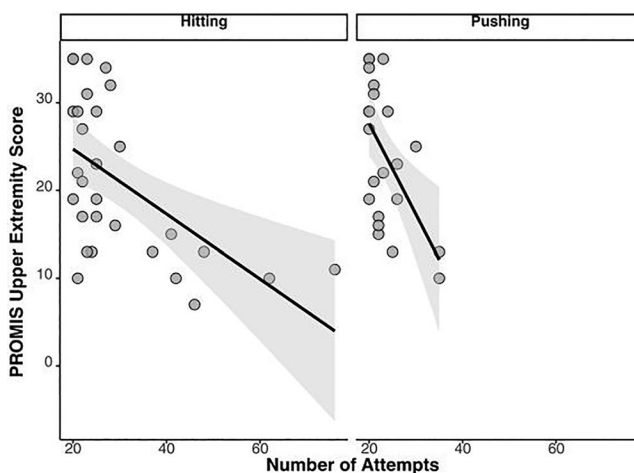


Figure 8. Association between task performance and self-reported upper-limb function (PROMIS Upper Extremity). In both task conditions (pushing and hitting), performance is indexed by the number of attempts required to complete 20 successful trials (higher values indicate poorer performance). Lines show linear fits with 95% confidence intervals.

MT was significantly associated with PROMIS Upper Extremity scores in either condition. Corresponding R^2 values were small (Pushing: .10 and .06 for mean and SD, respectively; Hitting: .004 and .001), indicating negligible explanatory value. One likely reason is that the task was self-paced: participants were not instructed to prioritize speed, nor were they placed under time pressure. Under these conditions, speed-accuracy tradeoffs are not well defined, and MT appears relatively insensitive to differences in upper limb function. Accordingly, MT was not adopted as a primary measure of task success for testing hypotheses concerning the relationship between V_{UCM} and performance.

Evidence of synergy

Descriptive statistics for ΔV indicated consistently positive values across all task conditions, limbs, and phases (see Figure 9). Mean ΔV values across limbs, conditions, and movement phases ranged from 0.71 to 0.80, with all observed values exceeding 0.43. This pattern indicates that variability was consistently greater within the uncontrolled manifold than its orthogonal complement, confirming the presence of task-stabilizing coordination. Given confirmation that the identified motor elements function synergistically in our experimental task, V_{UCM} can be used as an expression of use of motor abundance.

A linear mixed-effects model with participant as a random effect was used to examine how the Z-transformed ΔV (ΔV_Z) varied as a function of limb (paretic vs. non-paretic), task condition (hitting vs. pushing), and task phase (early vs. late). The model revealed a significant main effect of condition, $t(188.84) = -0.10$, $p = .004$. On average, ΔV_Z was lower in the pushing condition compared to the hitting condition. A significant main effect of limb was also observed, $t(188.84) = -0.20$, $p < .001$, indicating that the synergy index was lower when participants used their

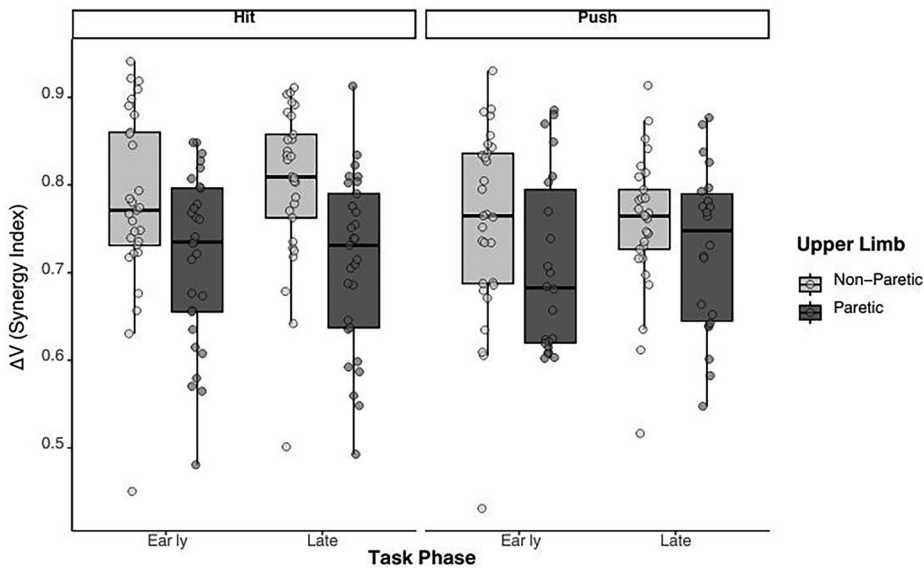


Figure 9. Mean z-transformed synergy index (ΔV_z) values across task condition (hitting vs. pushing), upper limb (paretic vs. non-paretic), and task phase (early vs. late). Positive ΔV_z values indicate greater variance along the uncontrolled manifold than in its orthogonal complement, consistent with the presence of motor synergies that stabilize task performance. Across all conditions, limbs, and phases, ΔV_z values were positive, suggesting that participants coordinated motor elements to support goal-directed action.

paretic limb compared to their non-paretic limb. These effects were qualified by a significant limb $\times$ condition interaction, $t(190.95) = 0.13$, $p = .02$. Post-hoc comparisons with Tukey adjustment revealed a significant difference between limbs in the hitting condition ($M_{\text{diff}} = 0.20$, 95% CI [0.11, 0.29], $p < .0001$), but not in the pushing condition ($M_{\text{diff}} = 0.07$, 95% CI [-0.03, 0.17], $p = .22$). There was no effect of movement phase.

This pattern closely mirrors the performance findings, suggesting that factors contributing to greater synergy strength—such as higher V_{UCM} magnitude reflecting greater use of motor abundance—may be associated with improved performance, after controlling for V_{ORT} as hypothesized.

Hypothesis testing

We conducted a series of targeted analyses focused on the relationships among use of motor abundance, gaze behavior, and upper limb performance in individuals post-stroke. Analyses were organized around the two specific aims described earlier.

Aim 1: relationship between use of motor abundance and upper limb performance

To test whether use of motor abundance predicts task performance, we fit separate linear mixed-effects models for the paretic and non-paretic limbs, each including participant as a random effect. The dependent variable was task performance, defined as the number of attempts required to complete 20 successful trials. Fixed effects

included V_{UCM} (log-transformed), V_{ORT} (log-transformed), task condition (dummy coded: hitting = 0, pushing = 1), movement phase (early = 0, late = 1), and all two- and three-way interactions among these predictors.

Because V_{UCM} and V_{ORT} were log-transformed, we multiplied their model coefficients (Tables 2 and 3) by $\log(1.10)$ to express effect sizes as the estimated change in performance associated with a 10% increase in each predictor on its original scale. These transformed values provide an interpretable approximation of how changes in use of motor abundance (V_{UCM}) and nonfunctional variability (V_{ORT}) are expected to influence task performance.

Paretic Limb. The model revealed a significant effect of V_{UCM} on task performance with the paretic limb, $t(77.09) = -3.71$, $p < .001$. As hypothesized, greater use of motor abundance was associated with better performance. According to the model, a 10% increase in V_{UCM} was associated with a reduction of approximately 0.44 attempts needed to complete the task, holding other predictors constant (see Table 2). This finding supports the hypothesis that greater use of motor abundance facilitates more efficient task execution after stroke.

V_{ORT} was also a significant predictor, $t(77.22) = 5.87$, $p < .001$: a 10% increase in V_{ORT} was associated with an estimated increase of approximately 0.76 attempts, indicating that greater nonfunctional variability was associated with poorer performance. This effect, however, was moderated by task condition, $t(73.65) = -5.70$, $p < .001$. In the pushing condition, the detrimental effect of V_{ORT} on performance was attenuated: a 10% increase in V_{ORT} was associated with an increase of approximately 0.37 attempts, compared to 0.76 attempts in the hitting condition. This

Table 2. Aim 1: Paretic limb mixed effect model examining the relationship between V_{UCM} and number of attempts to achieve 20 successful trials.

Predictor	Estimate	SE	<i>p</i>
Fixed effects			
Intercept	42.10	3.03	<.001
V_{UCM}	-4.60	1.24	<.001
V_{ORT}	8.02	1.37	<.001
Condition (Pushing)	-7.60	1.33	<.001
$V_{ORT} \times \text{Condition (Pushing)}$	-4.16	1.18	<.001
Random effect			
	Variance	SD	
Intercept (participant)	125.25	11.19	
Residual	5.89	2.43	

Note: V_{UCM} & V_{ORT} were log-transformed; therefore, model coefficients represent the change in performance associated with a proportional change in each predictor.

Table 3. Aim 1: Non-paretic limb mixed effect model examining the relationship between V_{UCM} and number of attempts to achieve 20 successful trials.

Predictor	Estimate	SE	<i>p</i>
Fixed effects			
Intercept	19.32	1.69	<.001
V_{UCM}	4.28	1.25	<.001
V_{ORT}	-1.39	0.94	>.14
Condition (Pushing)	1.66	0.95	>.09
$V_{UCM} \times V_{ORT}$	2.63	0.79	.001
$V_{UCM} \times \text{Condition}$	-2.22	0.83	.009
Random effect			
	Variance	SD	
Intercept (participant)	4.10	2.03	
Residual	3.33	1.83	

Table 4. Aim 2: Multivariate model examining the relationship between V_{UCM} and summary measures of head-gaze angle time series: mean, range and (log-transformed) fluctuation.

Predictor	Estimate	SE	<i>p</i>
Fixed effects			
Intercept	0.42	0.16	.01
Mean	0.01	0.004	.07
Range	0.02	0.004	<.001
Fluctuation	0.11	0.05	.02
Condition (Pushing)	0.10	0.06	.11
Limb (Non-paretic)	−0.15	0.05	.004
Phase (Late)	0.16	0.05	<0.001
Random effect			
	Variance	SD	
Intercept (participant)	0.07	0.26	
Residual	0.11	0.34	

finding suggests that greater tolerance for nonfunctional variability may be afforded when individuals are allowed to guide the puck across the bridge rather than having to release it early.

Non-paretic limb. For the non-paretic limb, the model revealed a significant main effect of V_{UCM} on task performance, $t(113.94) = 3.43$, $p < .001$. Contrary to our hypothesis, greater use of motor abundance was associated with worse performance. A 10% increase in V_{UCM} was associated with an estimated increase of 0.41 attempts needed to complete the task, holding other predictors constant. This effect, however, was moderated by task condition, $t(95.65) = -2.67$, $p = .009$. In the pushing condition, the performance cost associated with a 10% increase in V_{UCM} was reduced by approximately 0.20 attempts relative to the hitting condition.

A significant $V_{UCM} \times V_{ORT}$ interaction was also observed, $t(111.45) = 3.33$, $p = .001$. To visualize this interaction, we generated model-based predictions of task performance in the hitting condition as a function of standardized log-transformed V_{UCM} , evaluated at three representative levels standardized log-transformed V_{ORT} (−1 SD, mean, +1 SD), with other model terms held constant. As shown in [Figure 10](#), when V_{ORT} was low (1 SD below the mean), greater use of motor abundance (i.e. higher V_{UCM}) was associated with improved task performance, consistent with the pattern observed for the paretic limb. However, at average and high levels of V_{ORT} , the relationship between V_{UCM} and performance reversed—indicating that, in these situations, greater use of motor abundance was associated with worse task performance.

Aim 2: relationship between gaze behavior & use of motor abundance

We employed a multivariate mixed-effects model to examine the associations between motor abundance and three distinct gaze pattern variables. In this analysis, motor abundance served as the outcome variable, and head-gaze angle, gaze range, and gaze fluctuation were treated as predictors. Task condition (dummy coded: hitting = 0, pushing = 1), limb (dummy coded: non-paretic = 0, paretic = 1), and movement phase (dummy coded: early = 0, late = 1) were included as fixed effects, and participant was treated as a random effect. The analysis was restricted to trials in which the task variable was defined in three dimensions.

This modeling approach allowed us to assess the unique associations between motor abundance and multiple aspects of gaze behavior while accounting for shared variance

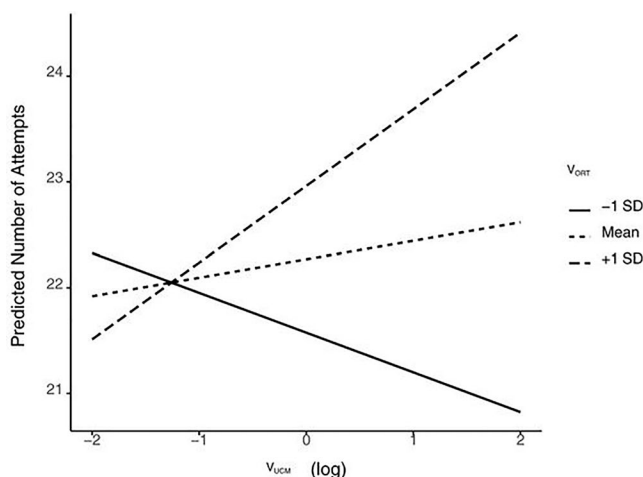


Figure 10. Predicted number of attempts as a function of use of motor abundance (V_{UCM}) for the non-paretic limb, evaluated at low (-1 SD), mean, and high ($+1$ SD) levels of nonfunctional variability (V_{ORT}). Both variables are standardized (z-scored) log-transformed values. Higher motor abundance was associated with better performance when V_{ORT} was low, but this relationship was attenuated and reversed at higher levels of V_{ORT} , suggesting that elevated nonfunctional variability may undermine the functional benefits of motor abundance.

among the gaze measures. Although motor abundance is specified as the dependent variable in the statistical model, the analysis is interpreted in relational terms, consistent with the view that gaze behavior and motor coordination are coupled processes within the ongoing perception-action cycle.

Results indicated that gaze range and gaze fluctuation were significantly associated with motor abundance when all three gaze variables were included in the model (see Table 4). Specifically, each one-degree increase in gaze range was associated with an approximate 1.7% increase in motor abundance, $t(216.72) = 3.97$, $p < .001$. For gaze fluctuation (log-transformed predictor), a 10% increase in fluctuation was associated with an estimated 1.1% increase in motor abundance, $t(203.88) = 2.36$, $p = .02$. The association between mean head-gaze angle and motor abundance was marginal, $t(159.78) = 1.83$, $p = .070$. Model estimates indicated that each one-degree increase in average gaze angle corresponded to an approximate 0.7% increase in motor abundance, suggesting a tendency for greater orientation toward task-relevant regions of the environment to be associated with increased use of motor abundance. These associations were observed while controlling for task condition, limb, and movement phase. Use of motor abundance was lower for the non-dominant limb compared to the dominant limb, $t(205.73) = -2.91$, $p = .004$, and greater during the later movement phase than the early phase, $t(187.23) = 3.58$, $p < .001$. The effect of task condition was not statistically significant, $t(209.29) = 1.59$, $p = .11$.

Overall, these findings indicate that multiple aspects of gaze behavior—specifically, greater orientation toward task-relevant regions of the environment, wider spatial exploration, and more dynamically structured reorientation of gaze—are uniquely associated with the use of motor abundance during task performance (see Figure 11). Within the perception-action cycle, this pattern is consistent with the idea that motor

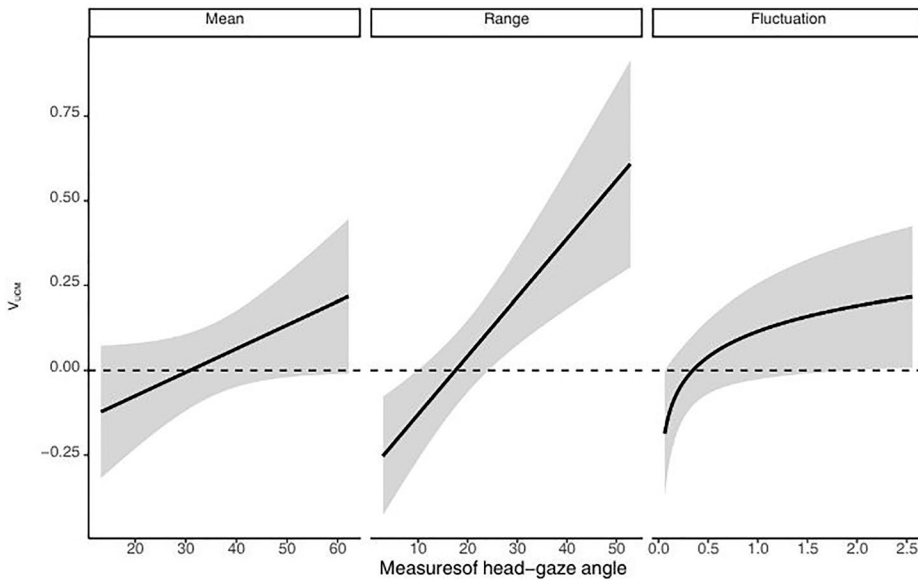


Figure 11. Association between motor abundance (V_{UCM}) and three measures of head-gaze angle: mean, range, and fluctuation. Lines represent model-based predictions from the multivariate mixed-effects model, and shaded regions indicate 95% confidence intervals derived from the fixed effects. Predictions are shown across the observed range of each gaze variable while holding the remaining gaze variables at their sample means and averaging across the observed distribution of hand, condition, and movement phase. Values on the y-axis are centered such that zero represents the sample mean of V_{UCM} ; the dashed horizontal line indicates average motor abundance. Gaze fluctuation (log-transformed in the model) is displayed on its original scale.

abundance supports the coordination required to both stabilize task-relevant variables and flexibly organize visual activity between the moving limb and task-relevant regions of the environment under evolving constraints.

Discussion

This study investigated how the use of motor abundance contributes to upper limb performance in individuals post-stroke, and whether gaze behavior helps explain individual differences in the use of motor abundance. Using a goal-directed object transportation task, we tested the hypothesis that greater use of motor abundance—operationalized as variability within the uncontrolled manifold (V_{UCM})—would predict better task performance. We then examined whether individual differences in gaze behavior were associated with the expression of motor abundance, consistent with the idea that perceptual engagement with the task environment underlies task-specific variations in motor behavior in response to prevailing conditions.

On the relation between use of motor abundance and performance

In line with our hypothesis, greater expression of motor abundance significantly predicted performance with the paretic limb: individuals who expressed greater functional

variability (V_{UCM}) required fewer attempts to complete the task. This relationship held across task conditions and was independent of nonfunctional variability (V_{ORT}). This finding provides empirical support for the claim that movement variability in individuals with stroke is not solely detrimental. Rather, overall movement variability includes a functional component that reflects the capacity to explore and stabilize novel motor solutions that create new forms of fit between a constrained movement repertoire and environmental features often not designed to accommodate disabled bodies.

To contextualize the practical significance of this effect, we examined the observed range of use of motor abundance in our sample. Specifically, a shift from -1 SD to $+1$ SD in log-transformed V_{UCM} corresponded to an approximately $2.5\times$ difference in motor abundance between individuals at the lower and upper ends of the distribution. According to model estimates (Table 2), this range translates to a difference of approximately 4.17 task attempts to achieve 20 successful trials. This difference reflects a considerable impact, underscoring the functional significance of exploring motor abundance when performing tasks with the impaired limb after stroke.

As expected, higher V_{ORT} —variability orthogonal to the uncontrolled manifold—was reliably associated with poorer task performance. This effect was more pronounced when participants used the *hitting* strategy for object transportation compared to the *pushing* strategy. There was also a large range of V_{ORT} in our sample: a shift from -1 SD to $+1$ SD in log-transformed V_{ORT} corresponded to a 169.9% increase in nonfunctional variability between individuals at opposite ends of the distribution. This was associated with a predicted increase of 7.96 attempts in the *hitting* condition and 3.83 attempts in the *pushing* condition. These results suggest that certain execution strategies may offer individuals more opportunity to compensate for or suppress disruptive variability—potentially by extending the time available for online corrections or by simplifying the control demands of the movement.

Together, these findings from participants' performance with the paretic limb highlight the importance of simultaneously amplifying the use of motor abundance and minimizing nonfunctional variability to support more effective upper-limb function. They also underscore the value of designing environments and tools that accommodate a range of execution strategies for everyday tasks. Doing so may offer individuals with disability greater opportunities to exploit their motor abundance and express effective movement solutions with greater ease, while also providing the necessary support to mitigate the negative impact of disruptive motor variability.

For the non-paretic limb, the relationship between use of motor abundance and task performance was moderated by task condition and the amount of nonfunctional variability. When V_{ORT} was controlled for—held at its average value in the sample—greater use of motor abundance was associated with poorer performance, particularly in the *hitting* condition. There was, however, a significant interaction between V_{UCM} and V_{ORT} . When V_{ORT} was low, greater use of motor abundance was associated with better performance, consistent with the pattern observed for the paretic limb. At average and high levels of V_{ORT} , this relationship reversed: increased use of motor abundance predicted worse performance. Notably, the magnitude of these effects was in the same range as those reported for the paretic limb.

These results suggest that there is a potential cost to the use of motor abundance (Gray, 2020). While all solutions within the UCM space are functionally equivalent

with respect to the controlled variable, they are unlikely to be equivalent in other respects (e.g. stability, effort, or tolerance to execution error)—see Wilson et al. (2016), for an example in the context of throwing. The implication is that some solutions may be more susceptible to stochastic influences (or more sensitive to perturbations), making them less reliable in practice. This would explain why individuals gravitate to a relatively smaller set of solutions at late stages of practice—at least in the context of simple experimental tasks (Latash, 2010). A critical question, then, is: under what conditions is the exploitation of motor abundance cost-effective?

We propose that exploitation of motor abundance is cost-effective (that is, supports performance) when the task imposes substantial demand relative to the performer's available solutions in a given context. Otherwise, it can result in performance loss. Our results provide initial support for this idea. Although task demands were held constant, the space of available motor solutions was narrower for the paretic limb due to limitations in motor repertoire. In this more constrained context, higher V_{UCM} likely reflected an adaptive response that allowed individuals to stabilize performance through greater exploration of the available motor solutions. For the non-paretic limb, however, the same task presented fewer constraints relative to movement capacity. In that context, when extensive exploration of motor abundance resulted in use of more error-prone strategies—reflected in relatively high V_{ORT} —there was a cost to performance.

Our proposed explanation for the complex relationship between use of motor abundance and performance should be explored in future studies. A promising direction would be to systematically manipulate the difficulty of the task relative to an individual's movement capacity—for instance, by imposing additional physical constraints or altering the precision requirements of the task—and assess whether the functional role of use of motor abundance shifts accordingly. Such work could help identify the boundary conditions under which exploration of motor abundance supports or undermines performance.

On the relation between gaze behavior and use of motor abundance

Our findings support the hypothesis that gaze behavior is systematically related to the use of motor abundance after stroke. Our measures of gaze behavior quantified the dynamic spatial relationship between head orientation and end-effector position during the course of the movement. Accordingly, small angles mean that head orientation remains closely aligned with the effector, whereas larger angles reflect visual exploration of more distal aspects of the task environment. Building on this, the range and fluctuation of this angle over time capture how much and how frequently/fast, respectively, this head $\times$ end-effector relationship changes during the course of the trial.

With this framework, our findings indicated that participants who, on average, looked farther ahead from the end-effector into the environment, varied the relative orientation between head and end-effector across a wider range, and showed more frequent and faster changes in that relationship over time, exhibited higher values of V_{UCM} . Importantly, these associations were observed within a multivariate model that accounted for shared variance among the different measures of gaze behavior, indicating that each aspect contributes uniquely to the expression of motor abundance.

These results are consistent with the study's hypothesis that motor abundance may support not only the stabilization of task-relevant variables but also the coordination required to sustain effective visual activity under changing task constraints. In the context of stroke—where visual impairments, altered postural control, and a tendency to anchor gaze to the body may increase the demands of informing sampling—this dual role may be particularly important for successful task performance. More broadly results give expression to Gibson's claim that visual perception is an embodied activity, realized through coordinated changes across the body that enable effective sampling of the optic array. In this sense, the same degrees of freedom that contribute to stabilizing task-relevant variables may also support the reorientation of the perceptual system toward informative regions of the environment.

The present findings establish a systematic relation between motor abundance and gaze behavior, but do not resolve their temporal organization. Changes in gaze patterns may support changes in motor abundance by enabling more effective sampling of task-relevant information, motor abundance may facilitate more effective gaze patterns by allowing flexible reorganization of the body, or both may evolve together as part of an ongoing perception-action cycle in support of task performance. Addressing these possibilities will require designs that track the temporal coupling between motor abundance and gaze behavior across practice over repeated trials, allowing for a more precise characterization of the perceptual-motor processes that support task performance and, in turn, may inform the development of more effective interventions to enhance functional recovery after stroke.

Limitations

The results of this study should be considered with respect to its limitations. First, the sample included individuals with a broad range of upper limb impairment severity, as reflected by PROMIS Upper Extremity scores ranging from 7 to 35 (out of a possible maximum of 35). Just as we observed differences in the relationship between V_{UCM} and performance across paretic and non-paretic limbs, it is possible that impairment severity moderates this relationship. Future studies should test whether the benefits of greater use of motor abundance differ across levels of motor impairment.

Second, in healthy individuals, eye and head movements are tightly coordinated during gaze shifts, with the two systems working together to direct visual attention across the environment (Franchak et al., 2021). However, how these coordination patterns are affected by stroke remains poorly understood. This study did not include detailed assessments of visual function, and it is possible that individuals with subtle or undiagnosed impairments—such as eye movement disorders or difficulties with binocular coordination—were included in the sample. As a result, our reliance on head-gaze angle may have provided an incomplete account of participants' gaze behavior. The integration of direct eye-tracking with head-based measures in future studies could offer a more comprehensive view of gaze behavior post-stroke and improve the interpretability of exploration-related findings.

Third, in the present study, hand position was selected as the task-level variable for quantifying motor abundance during task execution. While this choice captures how coordination stabilizes a performance-relevant variable, it does not fully characterize how movement and forces are ultimately transferred to the object, particularly

in the hitting condition. In this condition, successful performance depends on the interaction between the hand and the puck at the moment of contact, such that task outcome is only indirectly related to the stabilization of the hand itself. As a result, hand-based measures of motor abundance may not fully reflect the coordination processes that determine object motion following release. Object-level kinematic variables—such as puck velocity and angle of release—would provide a more direct link between motor coordination and task outcome and would complement the present approach. Future work should incorporate such measures to better capture how coordination at the level of the body gives rise to functional outcomes at the level of the object, particularly in tasks involving ballistic or contact-based actions.

Finally, limitations related to sample diversity should be acknowledged. Despite efforts to recruit a diverse participant pool, only 17% of included participants self-identified as Black, none self-identified as Hispanic or Latino, and the only Asian participant was excluded due to motor impairments that precluded task completion. This underrepresentation is particularly concerning given that stroke disproportionately affects racial and ethnic minority groups (Ikeme et al., 2022), yet these groups remain consistently underrepresented in stroke research. The lack of diversity limits the generalizability of our findings and underscores the need for more inclusive research efforts to inform the development of effective, culturally sensitive rehabilitation interventions.

Clinical implications

The present findings have important implications for the design and evaluation of rehabilitation strategies for individuals with stroke-related upper limb impairments. A central theme emerging from this work is that functional recovery need not to be defined—or constrained—by a return to able-bodied movement norms. Instead, our results suggest that promoting the exploration of available motor solutions, even if they deviate from canonical joint coordination patterns, can enhance performance and enable individuals to stabilize task-relevant outcomes despite persistent impairments.

In addition, our findings suggest that flexible movement coordination and visual engagement with the task environment are jointly organized during ongoing task performance. In this sense, visual activity emerges from the coordinated organization of the body, rather than from eye movements alone. Rehabilitation approaches that focus narrowly on retraining eye movements or visual scanning behavior—often through structured search task or exercises aimed at improving detection of stimuli across the visual field (Hanna et al., 2017)—may therefore overlook the critical, dual role that adaptive coordination of body movement plays in both stabilizing task-relevant variables and supporting effective sampling of visual information. Our results suggest that interventions should instead more explicitly integrate action coordination and visual activity during performance of meaningful tasks, rather than targeting each in isolation. One example would be to encourage orientation of gaze toward task-relevant regions of the environment while allowing flexibility in how movements are organized to achieve task goals. This integrated focus may help promote the emergence of more adaptive and functional behavior in individuals post-stroke. Future clinical studies could examine whether interventions of this kind are more effective in enhancing functional recovery than standard approaches.

Conclusion

This study showed that the relationship between the use of motor abundance and task performance post-stroke is complex and shaped by the interplay of motor impairment, nonfunctional variability, and task demands. When participants performed with the paretic limb—a condition that restricts movement repertoire—greater use of motor abundance predicted better performance regardless of task strategy. However, when participants performed with the non-paretic limb, the benefits of using motor abundance were conditional: performance gains with increased use emerged only when nonfunctional variability was low, with performance costs becoming apparent when exploratory flexibility coincided with relatively higher disruptive variability. Future studies should identify the boundary conditions under which the use of motor abundance supports or undermines performance.

Crucially, the use of motor abundance was systematically related to gaze behavior. Individuals who oriented their gaze toward task-relevant regions of the environment, explored a wider spatial range, and more dynamically reoriented their head relative to the end-effector exhibited greater motor abundance. These findings support the idea that motor abundance serves a dual role: stabilizing task-relevant variables while simultaneously supporting the coordination required for effective visual activity. In this sense, visual activity is not reducible to eye movements alone, but emerges from the coordinated organization of the body as a whole. Taken together, results suggest functional recovery after stroke may benefit from interventions that focus on enhancing movement possibilities—rather than on the normalization of movement patterns. Such interventions may better enable stroke survivors to engage with task-relevant information and more reliably and effectively achieve functional outcomes in everyday contexts.

Disclosure statement

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Data availability statement

Data processing code, statistical analysis code, and sample data are available on the Open Science Framework (OSF): <https://osf.io/2zx4m/>.

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